Reform, Rails and Rice: Thailand's Political Railroads and Economic Development in the 20th Century

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(1) Summaries

(1.1) Summary of research objective

- This paper examines the **impacts** of **politically-driven railway expansions** on **population** and **rice production**, particularly in the northern and northeastern regions of Thailand.

(1.2) Summary of background

- The earliest main routes were constructed in 1892, 1898 and 1900, connecting Bangkok to the **northeast, north** and **south**.
- These state-sponsored routes were targeted to **facilitate Bangkok's centralized administration**, but they were **not too extensive** to allow **France** and **Britain** to tactically use this infrastructure **for annexation**.
- These constructions have been named "**political railways**".
- **Ingram (1971)** claims that the **impact** on **rice cultivation** was **small**.
- On the contrary, **Kakizaki (2005)** argues that the north and northeast routes has significantly influenced the **expansion** of **trade** and **rice production**.

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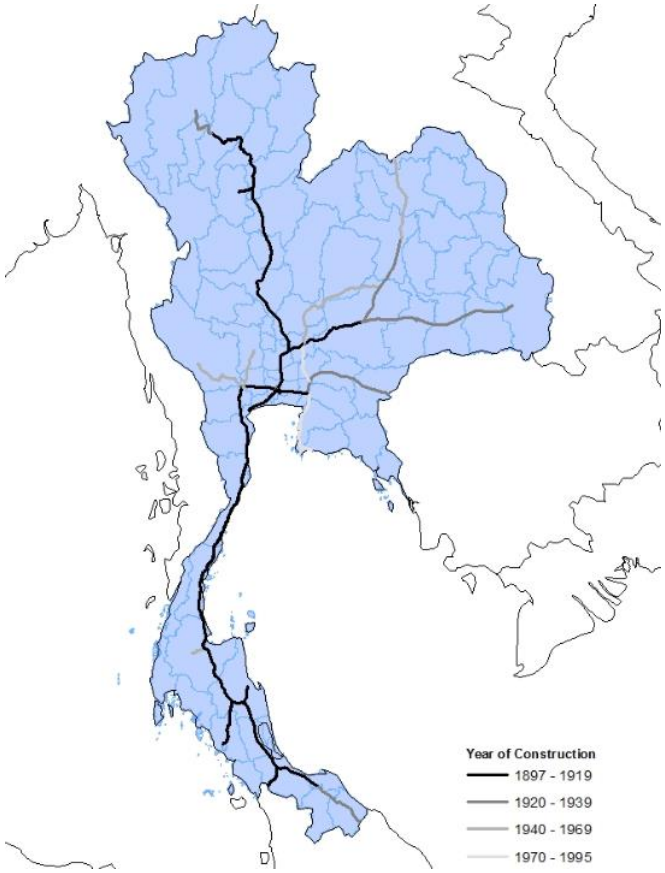
(1.3) Summary of data

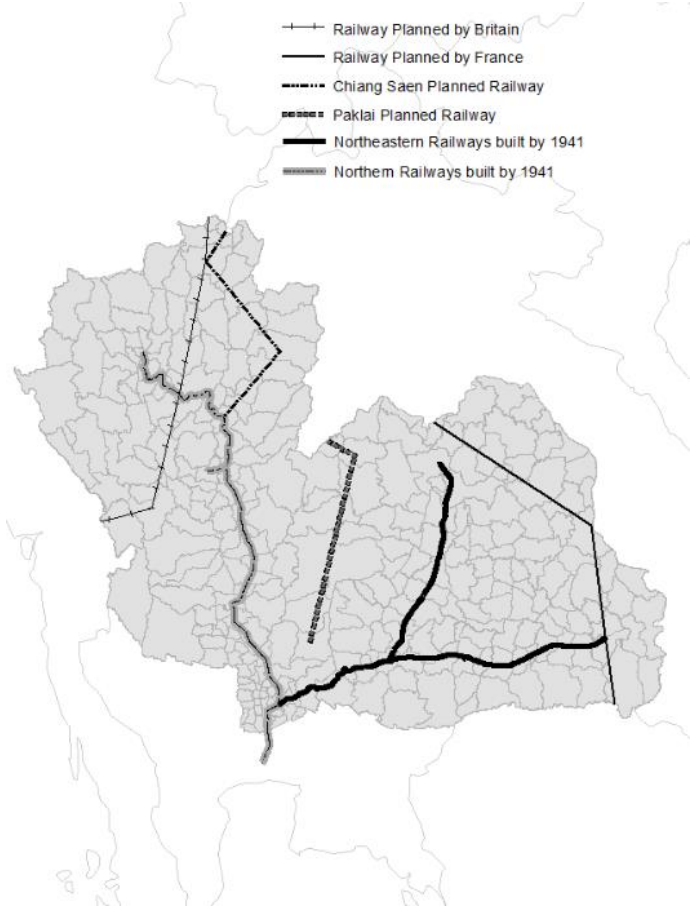
Maps of railway routes

Railway Network Expansion, 1897-1995



Source: Whyte (2010)

Planned and Completed Railways Lines, 1897-1941



Source: Kakizaki (2012)

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(1.3) Summary of data

Summary Statistics, 1947

Variable	Sources	Mean	Std. Dev.
District Population 1947 ('000)	1947 Census	49.58	28.17
Planted Area of Rice 1966 (Rai)	1947 Census	114,848	86,286
No Northern Rail Access	Whyte (2010); Ministry of Transport	0.87	0.34
No Northeastern Rail Access	Whyte (2010); Ministry of Transport	0.90	0.31
Dist to NE 1941 (Km)	Ministry of Transport	136.45	142.10
Dist to N 1941 (Km)	Ministry of Transport	130.85	143.40
Dist to Railway Planned by British	Kakizaki (2012); Ministry of Transport	256.45	183.75
Dist to Railway Planned by French	Kakizaki (2012); Ministry of Transport	240.27	149.88
Dist to Proposed Paklai Line (Km)	Kakizaki (2012); Ministry of Transport	157.90	97.27
Dist to Proposed Chiang Saen Line (Km)	Kakizaki (2012); Ministry of Transport	261.53	161.40
Longitude	Ministry of Transport	101.39	1.84
Latitude	Ministry of Transport	16.51	1.57
Area (Km2)	Ministry of Transport	1,582	1,286
Agricultural Suitability	Ramankutty et al. (2002)	0.35	0.14
Elevation Mean	Ministry of Transport	250.27	223.05
Elevation STD	Ministry of Transport	102.01	108.40
Distance to River (Km)	Ministry of Transport	3.58	9.04
Distance to Bangkok (Km)	Ministry of Transport	350.68	167.24
Observations	221		

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(1.3) Summary of data

Summary Statistics, 1966

Variable	Sources	Mean	Std. Dev.
District Population 1966 ('000)	1966 Provincial Statistical Year Books	60.78	37.38
Harvested Area of Rice 1966 (Rai)	1966 Provincial Statistical Year Books	102,012	108,405
No Northern Rail Access	Whyte (2010); Ministry of Transport	0.90	0.30
No Northeastern Rail Access	Whyte (2010); Ministry of Transport	0.90	0.30
Dist to NE 1941 (Km)	Ministry of Transport	137.47	143.98
Dist to N 1941 (Km)	Ministry of Transport	143.65	146.24
Dist to Railway Planned by British	Kakizaki (2012); Ministry of Transport	268.12	189.00
Dist to Railway Planned by French	Kakizaki (2012); Ministry of Transport	239.86	149.11
Dist to Proposed Paklai Line (Km)	Kakizaki (2012); Ministry of Transport	166.04	99.63
Dist to Proposed Chiang Saen Line (Km)	Kakizaki (2012); Ministry of Transport	271.87	163.31
Longitude	Ministry of Transport	101.47	1.88
Latitude	Ministry of Transport	16.49	1.56
Area (Km2)	Ministry of Transport	1,145	893
Agricultural Suitability	Ramankutty et al. (2002)	0.34	0.14
Elevation Mean	Ministry of Transport	252.62	217.40
Elevation STD	Ministry of Transport	98.52	105.05
Distance to River (Km)	Ministry of Transport	5.38	11.07
Distance to Bangkok (Km)	Ministry of Transport	357.91	165.93
Observations	310		

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(1.4) Summary of methodology

1.4.1 OLS specifications

Model #1

$$y = \beta_0 + \beta_1 NoRailN + \beta_2 NoRailNE + X'\gamma + \delta + \epsilon \quad (1)$$

Model #2

$$y = \beta_0 + \beta_1 NoRailN + \beta_2 DistRailN + \beta_3 NoRailNE + \beta_4 DistRailNE + X'\gamma + \delta + \epsilon \quad (2)$$

Model #3

$$y = \beta_0 + \beta_1 10kmRailN + \beta_2 20kmRailN + \beta_3 10kmRailNE + \beta_4 20kmRailNE + X'\gamma + \delta + \epsilon \quad (3)$$

1.4.2 Placebo tests

$$y = \beta_0 + \beta_1 NoRailN + \beta_2 DistRailN + \beta_3 NoRailNE + \beta_4 DistRailNE + \beta_5 DistToRailPlannedByBrit + \beta_6 DistToRailPlannedByFR + \varepsilon \quad \textbf{(4)}$$

$$y = \beta_0 + \beta_1 NoRailN + \beta_2 DistRailN + \beta_3 NoRailNE + \beta_4 DistRailNE + \beta_5 DistToPaklaiLine + \beta_6 DistToChiangSaenLine + \varepsilon \quad \textbf{(5)}$$

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(1.5) Summary of result

- The development of railways **increased** the area of **rice cultivation** and **population** in both cases of **1947** and **1966**.
- **The greater the proximity** to the railways, the higher numbers of **population** and **rice cultivation**.
- The **highway** development **attenuated** the magnitudes of coefficients in 1966.

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(2) Suggestions

(2.1) Suggestions on background

- **The roles of foreign advisors were significant in formulating policies.**

*"... The second **general advisor**, **Dr. E.H.Strobel**, took office in 1903. From that time to 1949, this post was reserved for an American, employed as an advisor in foreign affairs. Most American advisors were **Harvard-trained legal experts** who no affiliation with their own government. Strobel was followed in this position by **Jens I. Westengard**. W. H. Pitkin, Dr. Eldon James, Dr. Francis B. Sayre. Courtney Crocker, Raymond Stevens, R. Dolbeare, and Kenneth S. Patton ..."*

Source: Smith et al. (2005)

(2.1) Suggestions on background

*"… Traditionally, in **finance**, the advisor was **British**, beginning with E. Rivett-Carnac (1896-1903). Others were W.J.F. Williamson, Sir Edward Cook, and W. M. A. Doll. The British advisors urged the Thai government to follow a **cautious policy** and **build up reserves** in the Treasury. R L. Morant (1887-1893), W. G. Johnson, and E. S. Smith helped to develop Thai education, police, surveying and, and railways. **Danes** were employed in the navy and police, the **French** in law and public works, the **Italians** in architecture and construction work, and the **Germans** in the construction of the northern railway line …"*

Source: Smith et al. (2005)

(2.1) Suggestions on background

In 1907, the total number of foreigners employed in the Thai service was as follows :

British	126	Danish	39
French	5	Swedes	
American	4	Norwegians	
German	36	Japanese	9
Italian	12	Dutch	11
Belgian	5		

From British Foreign Office Papers 371/333 No. 9761 (Hereafter cited as F.O. 371/333)

Source: Numnonda(1971)

- The construction processes of **south, north and northeast** railways were **problematic**. All went through **delays** caused by **geographical constraints, financial limitations** and **lawsuits**.
- Some delays were caused by **conflicts** between the **German** general director of national railway department and the **British** construction contractor.

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ชุมทางบ้านภาชี – ลพบุรี	1 เมษายน 2444	43
ลพบุรี – ปากน้ำโพ (นครสวรรค์)	1 ตุลาคม 2448	117
ปากน้ำโพ – พิษณุโลก	24 มกราคม 2450	139
พิษณุโลก – ชุมทางบ้านดารา (อุตรดิตถ์)	11 พฤศจิกายน 2451[9]	69
ชุมทางบ้านดารา – สวรรคโลก (สุโขทัย)	15 สิงหาคม 2452	29
ชุมทางบ้านดารา – ปางต้นผึ้ง	15 สิงหาคม 2452	51
ปางต้นผึ้ง – แม่พวก	1 มิถุนายน 2454	19
แม่พวก – ปากปาน (แพร่)	15 พฤศจิกายน 2455	10
ปากปาน – ห้วยแม่ต้า	1 พฤษภาคม 2456	12
ห้วยแม่ต้า – บ้านปิน	15 มิถุนายน 2457[10]	13
บ้านปิน – ผาคอ	1 พฤษภาคม 2458	17
ผาคอ – แม่จาง	15 ธันวาคม 2458	19
แม่จาง – ลำปาง	1 เมษายน 2459	42
ลำปาง – ปางหัวพง	20 ธันวาคม 2459	33
ปางหัวพง – ปางยาง (ลำพูน)	1 กรกฎาคม 2461	4
ปางยาง – อุโมงค์ขุนตาน – เชียงใหม่	1 มกราคม 2469	72

1 April 1901

1 January 1926

ตารางที่ 1 สถานีรถไฟ วันที่เปิดให้บริการ และระยะทางของรถไฟสายเหนือ

Northern route: distance and starting date of operation

Source: www.tri.chula.ac.th/triresearch/north/north.html

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สถานีสำคัญ	วันที่เปิดบริการเดินรถไฟ	งระหว่างสถานี (โลเมตร)
กรุงเทพ – เพชรบุรี	1 เมษายน 2446	
เพชรบุรี – ชะอำ	9 มิถุนายน 2454	
ชะอำ – หัวหิน	25 พฤศจิกายน 2454	
หัวหิน – วังก์พง	1 มกราคม 2456	
วังก์พง – ประจวบคีรีขันธ์	1 มิถุนายน 2457	69
ประจวบคีรีขันธ์ – บ้านกรูด	1 มิถุนายน 2457	59
บ้านกรูด – บางสะพานใหญ่	1 ธันวาคม 2458	16
บางสะพานใหญ่ – ชุมพร	1 ตุลาคม 2459	
ทุ่งสง – บ้านนา	14 กุมภาพันธ์ 2458	
ทุ่งสง – พัทลุง	1 ตุลาคม 2457	89
พัทลุง – อู่ตะเภา[10] (ชุมทางหาดใหญ่)	1 มกราคม 2456	84
อู่ตะเภา – คลองทราย	1 เมษายน 2460	100
คลองทราย – บาลอ	1 พฤศจิกายน 2463	
บาลอ – ตันหยงมัส	1 มีนาคม 2463	38
ตันหยงมัส – สุไหงโกลก	17 กันยายน 2464	45
ชุมทางหาดใหญ่ – บาดังเบซาร์	1 กรกฎาคม 2461	

1 April 1903

17 September 1921

ตารางที่ 1 สถานีรถไฟ วันที่เปิดให้บริการสถานรถไฟสายใต้

Southern route: distance and starting date of operation

Source: www.tri.chula.ac.th/triresearch/south/south.html

สถานีสำคัญ	วันที่เปิดบริการ เดินรถไฟ	ถานีกรุงเทพ ...ลเมตร)
กรุงเทพฯ -อยุธยา	26 มีนาคม 2439[17]	71
สายกรุงเทพ – อุบลราชธานี		
ชุมทางแก่งคอย	1 พฤศจิกายน 2440[18]	125
ปากช่อง	25 พฤษภาคม 2442	
นครราชสีมา	21 ธันวาคม 2443	246
ท่าช้าง	1 พฤศจิกายน 2468	285
บุรีรัมย์	1 เมษายน 2468	376
สุรินทร์	1 พฤษภาคม 2469	420
ห้วยทับทัน	1 พฤษภาคม 2470	482
ศรีสะเกษ	1 สิงหาคม 2471	515
อุบลราชธานี	1 เมษายน 2473	575
สายกรุงเทพ – หนองคาย		
ชุมทางบัวใหญ่	1 พฤษภาคม 2474[19]	346
เมืองพล	2474	378
บ้านไผ่	2475	408
ขอนแก่น	1 เมษายน 2476	0
อุดรธานี	24 มิถุนายน 2484	9
หนองคาย	23 กันยายน 2499	624
ท่านาแล้ง	5 มีนาคม 2552	627

26 March 1896

26 March 1956

ตารางที่ 1 สถานีรถไฟ วันที่เปิดให้บริการ และระยะทางของรถไฟสายตะวันออกเฉียงเหนือ

Northeast route: distance and starting date of operation

Source: www.tri.chula.ac.th/triresearch/neast/neast.html

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(2.1) Suggestions on background

- **The fear of annexation also influenced the fiscal and monetary policies to maintain both fiscal surplus and current account surplus.**
- **Most infrastructure and R&D investments were constrained.**

*"… In 1926, **King Prajadhipok**, long recognizing the urgency of the economic situations, immediately moved to find effective solutions and to **quickly mend the huge deficits**. He created a Financial Advisory Committee to control the expenses; he also engaged a British financial advisor, **Sir Edward Cook** …"*

Source:
https://www.bic.moe.go.th/images/stories/UNESCO/Great_Personalities/1._His_Majesty_King_Prajadhipok_King_Rama_7_for_press.pdf

(2.1) Suggestions on background

"… King Prajadhipok's government was able to recover from the crisis in the first few years, for in the 1927 and 1928 the budgets were seen as greatly improved, with ***surplus funds*** *to spend on public works and with some* ***cash reserve*** *…"*

"… After that, the ***budget*** *seemed to continue to* ***stabilize for a few years****. Sir Edward Cook having completed his term of service,* ***left Thailand*** *with* ***praises for his success*** *…"*

Source:
https://www.bic.moe.go.th/images/stories/UNESCO/Great_Personalities/1._His_Majesty_King_Prajadhipok_King_Rama_7_for_press.pdf

(2.1) Suggestions on background

Fact #1: **Europeans** were **directly involved** in both processes of **planning** and **construction** of all railways.

Fact #2: Not only did the political purposes, but both **engineering limitation** and **geographical constraint** also **determined** the **selection of routes**.

Fact #3: A **fear of annexation** influenced the government to **maintain the fiscal surplus**, but the railway constructions **were mostly financed by foreign loans**.

(2.2) Suggestions on related literature

Listed below are studies examining the impacts of trains.

Japan: Sands (1993), Cervero & Bernick (1996), Banister & Berechman (2000), Urena et al. (2009), Murakani & Cervero (2012)

Spain: Gutierrez et al. (1996)

Germany: Ahlfeldt & Feddersen (2009)

UK: Freeman (2007)

Japan, France, Germany, Spain, and Italy: Albalate & Bel (2010)

(2.3) Suggestions on data

Some surveys could be the alternative sources of data of additional control variables.

- Report of the First Annual Exhibition of Agriculture and Commerce, 1910, by Barnett, J.C.
- Report of the Second Exhibition of Agriculture and Commerce, 1911, by Alexander Morning Ltd.
- Siam Rural Economic Survey, 1930-31, by Zimmerman, C.C
- Siam 2nd Rural Economic survey, 1934-35, by Andrews, J.M.

REPORT OF THE

FIRST ANNUAL EXHIBITION

OF

AGRICULTURE & COMMERCE.

HELD IN

BANGKOK, APRIL 1910.

BY

J. C. BARNETT.

Illustrated from Photographs by
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Half Tone Blocks by
J. ANTONIO, BANGKOK.

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in English and Siamese
by the Bangkok Daily Mail.

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SECOND EXHIBITION

OF

Agriculture & Commerce

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SIAM

2nd. RURAL ECONOMIC SURVEY

1934—1935.

By

JAMES M. ANDREWS, A. B., M. A.

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Harvard University.

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(2.3) Suggestions on methodologies

Many econometrical methods have been continuously developed and conventionally applied to examine the spillover of infrastructure.

(1) OLS regression

(2) 2SLS regression

(3) Panel regression

- Fixed effect
- Random effect

(4) Spatial regression

- Spatial lag model
- Spatial error model

(5) Spatial panel regression

(6) Spatial 2SLS regression

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Thank you very much

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